Lecture plan · for the lecturer

Bias, variance and cross-validation

Data science · University · Lecture 2

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Decompose expected error.
- Estimate error with cross-validation.
- Prevent leakage in student work.

Activity: Find the leak

1. Warm-up (5 min): Show two models' training and test errors and ask students to vote, then reveal the real-world results.
2. Teach (15 min): Derive the decomposition and fit polynomials of rising degree live.
3. Hunt (20 min): Pairs compare CV scores across the leaky and clean notebooks and find each leak.
4. Fix (15 min): Rewrite one notebook with a Pipeline and grouped folds, and compare the honest score.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Why does training error always fall?
- What is the unit of independence in your data?
- How will your brief stop leakage?

Exit ticket

- Name one leak and the line of code that fixes it.

Try it in your next lecture

Hand out two short notebooks, one with leakage and one without, and ask pairs to explain why their cross-validation scores differ before you say anything.

Seminar sheet · for students

Bias, variance and cross-validation

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Name _____ Class _____ Date _____

1 **A student standardises all the data, then runs 5-fold cross-validation. Is the estimate honest?**

◯ Yes, scaling doesn't use the labels ◯ No, the test folds leaked into the scaling

◯ Yes, if k is at least 5

2 **Work out the cross-validation estimate.**

Fold errors 10, 12, 8, 14, 11 give a 5-fold CV error of _____ . The final test set is used _____ times for choosing the model.

Word bank: 11 · 55 · 5 · 0

3 **Which of these leak information into a cross-validation estimate? Choose all that apply.**

☐ Selecting the 20 most correlated features on all the data, then running CV

☐ Several rows per student, split randomly across folds

☐ Fitting the scaler inside each training fold ☐ Filling missing values with the mean of all rows first

4 **Adding more training data mainly reduces a model's variance, not its bias.**

◯ True ◯ False

5 **Match each symptom to its most likely cause.**

Training and test error both high	• • • •	Too few data or folds
Training error low, test error high	• • • •	Leakage
CV score far better than the later real-world score	• • • •	High variance
CV error changes a lot between random splits	• • • •	High bias

✎ **Think of a student project you marked last year. Could leakage have inflated its score? How would your project brief prevent it now?**